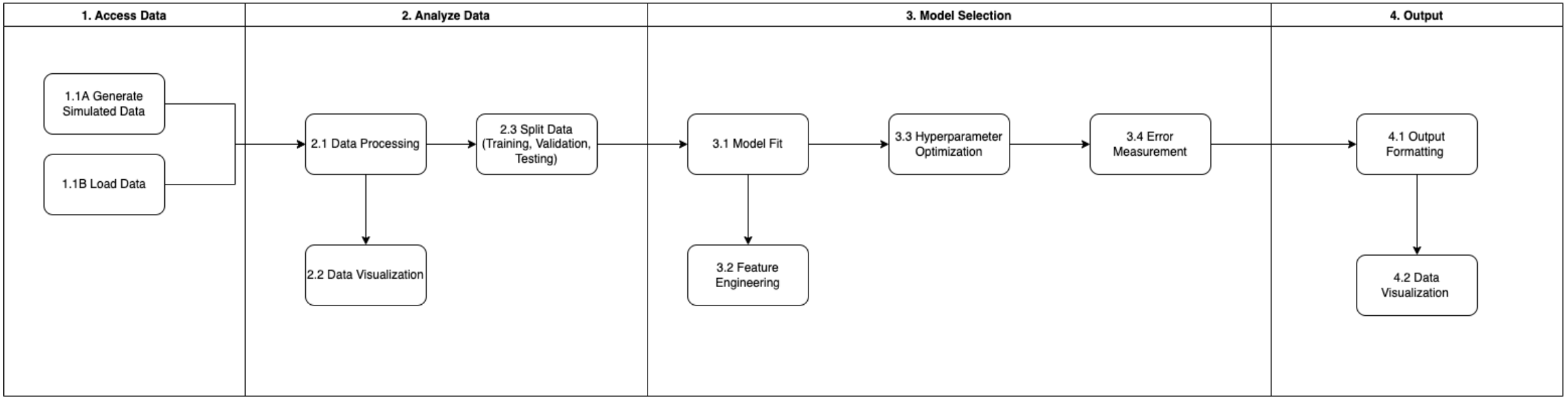
Ambient Systems

Model Documentation

Machine Learning Process



Part 1: Access Data

Part 1.1B: Load Data

- Source 1: equipment setting (timestamp, setting 1, 2, 3, 4, etc.)
- Source 2: sensor reading (timestamp, reading 1, 2, 3, 4, etc.)
- Source 3: user-defined optimal (timestamp, reading 1, 2, 3, 4, etc.)

Part 2: Analyze Data

Step 2.1: Data Processing

- Filter outliers and null values
- Group by regular intervals and take average among the readings

  each source of data with a name tag and zone tag
- Join Tables based on time of reading

Part 3: Model Selection

3.1: Model Fit (Regression)

- Regression 1: (X (Source 1, settings), Y (Source 2, sensors))
- Regression 2: (X (Source 3, preference), Y (Source 1, settings)) --> give settings to the user

3.2: Feature Engineering

- Identify which features appear correlated → add expert review

Part 3: Model Selection

3.3: Hyperparameter Optimization

MATLAB References

- Get Ready for AI with MATLAB
- MATLAB Join